

Modern Web Architecture

A 12-Week Curriculum for Secondary Students and Adult Learners

Author: Jeremy Ivy

Focus: HTML5 · CSS3 · JavaScript (Vanilla)

Alignment: IGCSE 0417 Practical **Approach:** Logic Before Tools
Requirements

Course Overview

This curriculum teaches students to build websites from the ground up using only HTML, CSS, and vanilla JavaScript. No frameworks, no dependencies, no shortcuts. Students learn to see causality: write a line of code, watch the page change. By Week 12, each student deploys a live portfolio site they built themselves.

Target Learners: Secondary students (ages 14–18) or adult learners with no prior coding experience. L2 English speakers supported throughout.

Prerequisite: Basic computer literacy. No prior programming experience required.

Curriculum Structure

Phase 1: The Skeleton (Weeks 1–4)

HTML Fundamentals

Document anatomy, semantic structure, data & media, hyperlinks & navigation. Students learn that HTML is meaning, not just markup.

Phase 2: The Skin (Weeks 5–8)

Pure CSS Mastery

Box model, typography & hierarchy, Flexbox layouts, CSS Grid. Students move from 'making it pretty' to 'designing information architecture.'

Phase 3: The Brain (Weeks 9–12)

Logic & Integration

CSS variables & DRY coding, JavaScript & the DOM, logic & state, deployment & production. Students own causality before they inherit abstractions.

Week-by-Week Map

Week	Topic	Core Skill	Deliverable
1	Semantic Structure	Meaningful markup	Accessible document
2	Data & Media	Lists, tables, figures	Structured content page
3	Hyperlinks & Navigation	Path logic	Multi-page mini-site
4	Document Anatomy	Boilerplate mastery	Standards-compliant template
5	The Box Model	Spatial reasoning	Layered component
6	Typography & Hierarchy	Visual rhythm	Type-system page
7	Flexbox Layouts	1D alignment	Navigation bar + card grid
8	CSS Grid Mastery	2D layout	Magazine-style layout
9	Variables & DRY Coding	System thinking	Theme-switchable page
10	JavaScript & The DOM	Interactivity	Click-responsive component
11	Logic & State	Decision structures	Conditional UI
12	Deployment & Production	Live publishing	Public portfolio site

IGCSE 0417 Alignment: Weeks 2, 4, and 7 directly address practical exam requirements. All weeks reinforce the computational thinking strands assessed in the written paper.

Sample Teacher's Guide: Week 1

Semantic Structure — Teaching 'Meaning' over 'Visuals'

1. The Hook: The Generic Grocery Store

Ask students to imagine a grocery store where every box is plain white with no label. You want milk, but you have to open every box to find it. A `<div>` is a plain white box. A `<header>` is a label. This helps the browser (and search engines) find what it needs without opening every box.

2. Conceptual Model: The 'Boss & Workers'

Personify the tags to help L2 learners understand hierarchy:

- **`<body>`** — The Office (where everything happens)
- **`<header>`** — The Boss (sets the name and tone)
- **`<nav>`** — The Map (shows the way to other rooms)
- **`<main>`** — The Real Work (the most important info)
- **`<footer>`** — The Assistant (small details at the bottom)

3. L2 Language Support

Aside: Call it 'Extra Info' or 'Side-note.' It's the 'P.S.' of a web page.

Section vs. Article: A Section is a chapter in a book; an Article is a post that can stand alone.

Start/Stop: Use instead of Opening/Closing. 'If you don't say STOP (`</h1>`), the computer thinks the whole page is a headline.'

4. Practical Activity: The Vandalism Hunt

Have students open a news site, right-click the main title, and select "Inspect." Change the text inside the `<h1>` to their own name. This proves that `<h1>` isn't just "big text"—it's the tag the computer looks for to find the Title.

Sample Teacher's Guide: Week 5

The Box Model — Visualizing Layout through Physical Space

1. The Hook: The Christmas Gift

Teach the four layers using the 'Present' analogy:

- **Content:** The Toy (the actual text or image)
- **Padding:** The Bubble Wrap (protection *inside* the box)
- **Border:** The Cardboard Box (the visible edge)
- **Margin:** The 'Keep Away' Distance (space *outside*, between this gift and others)

2. Pedagogy Technique: The 'Red Border' Trick

Students often get frustrated because they can't "see" where one element ends and another begins. For the first three weeks of CSS, have every student add this rule to their stylesheet:

```
* { border: 1px solid red; }
```

This makes the "invisible" boxes visible. They can see exactly how padding pushes content in and how margin pushes boxes apart.

3. L2 Language Support

Padding: Call it 'Inner Space.' It makes the box bigger from the inside.

Margin: Call it 'Personal Space.' It's like people standing 1 meter apart.

Border-Box: Call it 'The New Rule.' It tells the computer: 'The border and padding stay **INSIDE** the width I gave you.'

4. Practical Activity: The 'Button' Challenge

Have students create a simple link <a>. Their goal is to turn it into a "Button" by adding a background color and then using **Padding** and **Margin** to make it look clickable.

Success Criteria: The text should not touch the edge of the color (Padding), and the button should not touch the text next to it (Margin).

Sample Teacher's Guide: Week 12

Deployment & Production — From Local Files to Live URLs

1. The Hook: The Housewarming Party

Explain the difference between building a house (Local) and inviting people to see it (Live). Up until now, your website has been a secret on your laptop. **Deployment** is like having a housewarming party. You have to clean up the trash (delete unused files) and make sure the door (the URL) is easy for people to find.

2. Pedagogy Technique: Error Literacy

Students will panic when their site looks different on a live server than it did on their computer. Teach them to:

- Check that file names match exactly (servers are case-sensitive; cat.jpg ≠ Cat.jpg)
- Verify all paths are relative, not absolute (no C:/Users/...)
- Open the Browser Console (F12) and look for red error text before asking for help

3. L2 Language Support

Deployment: Call it 'Going Live.' Moving the code from your computer to the Internet.

Production: Call it 'Professional Mode.' No messy code, no unused images, everything is clean.

Optimization: Call it 'Making it Fast.' Smaller images and cleaner code so the user doesn't have to wait.

4. Practical Activity: The Launch Party

Use a free hosting service (GitHub Pages, Netlify, or Vercel) to publish each student's final portfolio. Every student sends their URL to a "Classroom Registry." Then, each student visits two other portfolios and provides one piece of positive feedback and one suggestion for improvement.

Success Criteria: Every student has a working, public link they can show to parents or future employers.

L2 Vocabulary Support

Three-phase glossaries designed for English language learners in Cambodian classrooms

Phase 1: The Skeleton (HTML)

Tag (*Label*) — The markers (< >) that tell the computer what a piece of text is.

Element (*Part*) — The whole 'sandwich': start tag, content, and end tag.

Attribute (*Extra Info*) — Words inside a tag that give more detail, like a link destination.

Nesting (*Inside / Outside*) — Putting one element inside another, like a box inside a bigger box.

Semantic (*Meaningful*) — Choosing a tag based on what the content is (e.g., <nav> for menus).

Path (*Address*) — The specific direction the browser follows to find a file.

Phase 2: The Skin (CSS)

Selector (*The Target*) — The name of the HTML part you want to style.

Declaration (*The Rule*) — The property and value together (e.g., color: blue;).

Padding (*Inner Space*) — Space inside the box, between text and border.

Margin (*Outer Space*) — Space outside the box. Pushes other elements away.

Hierarchy (*Level of Importance*) — Using size and color to show which text matters more.

Responsive (*Flexible*) — The website changes size to fit phones and computers.

Phase 3: The Brain (JavaScript)

Variable (*Storage Box*) — A named place to save information like a name or score.

Condition (*The Choice*) — A test to see if something is true or false.

Function (*A Task*) — A set of instructions the computer saves and runs when asked.

Event (*The Trigger*) — Something the user does, like clicking or scrolling.

The DOM (*Remote Control*) — The system JavaScript uses to change HTML and CSS.

State (*Current Status*) — What the app looks like right now (e.g., 'The menu is open').

About the Curriculum Designer

Jeremy Ivy is a frontend developer and ESL educator based in Cambodia. He spent two decades building interfaces for global retail and healthcare platforms before applying that discipline to classrooms where every kilobyte matters. He teaches logic before syntax and builds tools that respect the user's context.

Contact: jeremyteaches76@gmail.com

Portfolio: jeremyivy.com

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Full 12-week syllabus, student lessons, teacher guides, and glossaries available upon request.